Here are the detailed, easy-to-understand notes covering the second half of the "Solutions" chapter, focusing on Colligative Properties and Abnormal Molar Masses.

Chemistry Notes: Solutions (Part 2)

6. Colligative Properties and Determination of Molar Mass

When a non-volatile solute is added to a volatile solvent, the vapour pressure of the solvent decreases. Connected to this decrease are four specific properties known as **colligative properties** (from Latin *co* meaning together, *ligare* meaning to bind).

- **Definition:** Colligative properties are properties of solutions that depend strictly on the **number** (or concentration) of solute particles relative to the total number of particles, regardless of the chemical identity or nature of the solute.

These properties are incredibly useful for experimentally determining the molar mass of an unknown solute.

A. Relative Lowering of Vapour Pressure

As established by Raoult's Law, adding a non-volatile solute lowers the solvent's vapour pressure. The relative lowering of vapour pressure is perfectly equal to the mole fraction of the solute (x_2).

- **Formula:** $\frac{p_1^0 - p_1}{p_1^0} = x_2$
  (Where p_1^0 is the vapour pressure of pure solvent, and p_1 is the vapour pressure of the solution).
- **For Dilute Solutions:** If the solution is very dilute, the moles of solute (n_2) are much smaller than moles of solvent (n_1), so the denominator can be simplified to just n_1. The expanded formula used to find molar mass (M_2) is:
  $\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 \times M_1}{M_2 \times w_1}$
  (Where w stands for mass and M stands for molar mass).

B. Elevation of Boiling Point (ΔT_b)

A liquid boils when its vapour pressure becomes equal to the surrounding atmospheric pressure. Because a non-volatile solute *lowers* the vapour pressure, the solution must be heated to a **higher temperature** to make its vapour pressure reach atmospheric pressure.

- **Formula:** The elevation in boiling point (ΔT_b) is directly proportional to the **molality (m)** of the solution.
  $\Delta T_b = K_b \times m$
- **Constants:** K_b is known as the **Boiling Point Elevation Constant** or **Ebullioscopic Constant**. Its unit is $\mathrm{K\ kg\ mol^{-1}}$.

C. Depression of Freezing Point (ΔT_f)

A substance freezes when the vapour pressure of its liquid phase equals the vapour pressure of its solid phase. The lowered vapour pressure of the solution means it will hit the solid phase's vapour pressure curve at a **lower temperature**.

- **Formula:** The depression in freezing point (ΔT_f) is also directly proportional to the **molality (m)** of the solution.
  $\Delta T_f = K_f \times m$
- **Constants:** K_f is known as the **Freezing Point Depression Constant** or **Cryoscopic Constant** (unit: $\mathrm{K\ kg\ mol^{-1}}$).

D. Osmosis and Osmotic Pressure (Π or P)

Certain networks like pig's bladder, parchment, or cellophane act as **Semipermeable Membranes (SPM)**. They contain submicroscopic pores that allow small solvent molecules (like water) to pass through, but block larger solute molecules.

- **Osmosis:** The spontaneous flow of solvent molecules through a semipermeable membrane from a pure solvent (or a dilute solution) into a more concentrated solution.
- **Osmotic Pressure (P or Π):** The exact amount of excess mechanical pressure that must be applied to the solution side to precisely **stop** the flow of osmosis.
- **Formula:** For dilute solutions, osmotic pressure is directly proportional to the **Molarity (C)** of the solution at a given temperature (T).
  $P = CRT$ (where R is the gas constant).
- **Advantage for Biomolecules:** The osmotic pressure method is widely preferred for finding the molar mass of biomolecules (like proteins and polymers) because it is measured at **room temperature** (protecting fragile molecules from heat), uses molarity instead of molality, and produces a remarkably large, easy-to-measure pressure even in very dilute solutions.

Important Terms Related to Osmosis:

- **Isotonic Solutions:** Two solutions with the exact same osmotic pressure. No osmosis occurs between them. (e.g., 0.9% NaCl solution is isotonic with blood cells).
- **Hypertonic Solutions:** A solution with a *higher* concentration. If a blood cell is placed in it, water flows *out* of the cell, causing it to shrink.
- **Hypotonic Solutions:** A solution with a *lower* concentration. If a cell is placed in it, water flows *into* the cell, causing it to swell.
- **Everyday Examples:** Wilted flowers revive in water; eating too much salt causes water retention in tissue cells leading to puffiness (**edema**); and salting meat preserves it by pulling water out of bacteria, killing them.

E. Reverse Osmosis and Water Purification

If a pressure **larger** than the osmotic pressure is applied to the solution side, the process goes backward. Pure solvent is actively squeezed out of the solution.

- **Application:** Reverse osmosis is widely used for the **desalination of sea water** to produce pure drinking water. A strong film of **cellulose acetate** is typically used as the membrane.

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7. Abnormal Molar Masses and van't Hoff Factor

Colligative properties depend on the *number* of particles. If a solute changes its particle count when dissolved, the calculated molar mass will be wrong—this is called an **abnormal molar mass**.

A. Dissociation and Association

- **Dissociation (Breaking apart):** Strong electrolytes like KCl break into ions (K^+ and Cl^-) in water. One mole of solute becomes *two* moles of particles. This doubles the colligative effect, resulting in an experimentally determined molar mass that is **lower** than the true value.
- **Association (Clumping together):** Molecules like ethanoic acid (acetic acid) hydrogen bond and pair up (dimerise) in solvents like benzene. Two moles of molecules clump into *one* mole of particles. This halves the colligative effect, making the calculated molar mass **higher** (twice the expected value).

B. The van't Hoff Factor (i)

In 1880, van't Hoff introduced a correction multiplier, i, to account for the exact extent to which a solute associates or dissociates.

- **Definition:**
  $i = \frac{\text{Normal molar mass}}{\text{Abnormal (Observed) molar mass}}$
  $i = \frac{\text{Observed colligative property}}{\text{Calculated colligative property}}$
  $i = \frac{\text{Total moles of particles AFTER association/dissociation}}{\text{Total moles of particles BEFORE association/dissociation}}$
- **Interpreting i:**
  - For **association**: $i < 1$ (e.g., $i \approx 0.5$ for ethanoic acid in benzene).
  - For **dissociation**: $i > 1$ (e.g., $i \approx 2$ for NaCl).

C. Modified Equations

By plugging the van't Hoff factor into our colligative property equations, we fix the particle count and get perfectly accurate answers.

1. Relative lowering of vapour pressure: $\frac{\Delta p_1}{p_1^0} = ix_2$ (approx. $\frac{in_2}{n_1}$)
2. Elevation of Boiling Point: $\Delta T_b = iK_b m$
3. Depression of Freezing Point: $\Delta T_f = iK_f m$
4. Osmotic Pressure: $P = iCRT$